BEEVIL KNIEVEL: Precision Edge-AI Apiculture Monitoring Platform

IEEE HardwAIre Challenge 2026 Phase 2 — Autonomous In-Hive Acoustic & Microclimate Telemetry

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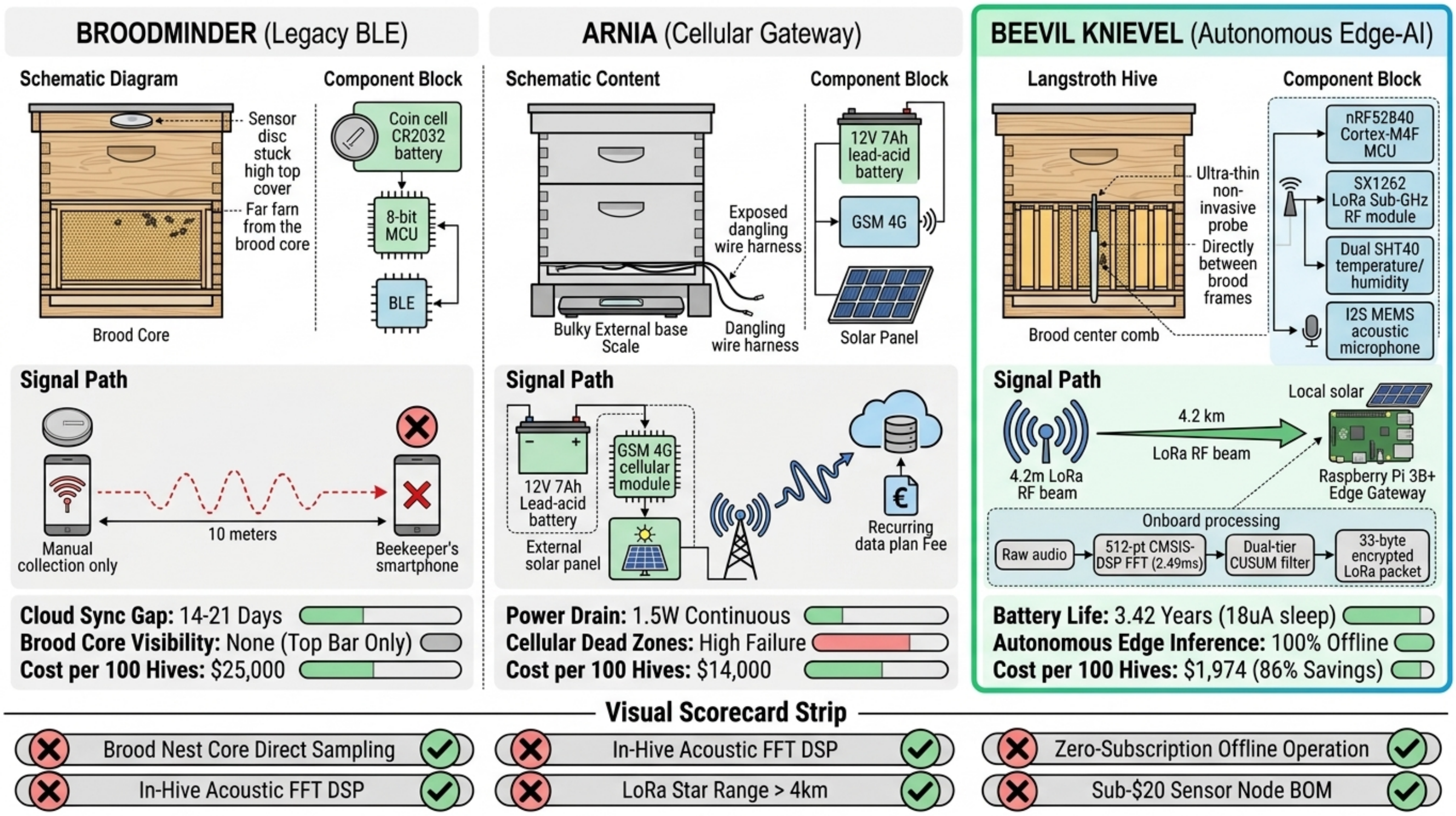

Figure 1: Architectural Comparison — Legacy BLE (BroodMinder) vs. Cloud Cellular (Arnia) vs. Autonomous Edge-AI (BEEVIL KNIEVEL). Direct in-comb probe vs. frame-

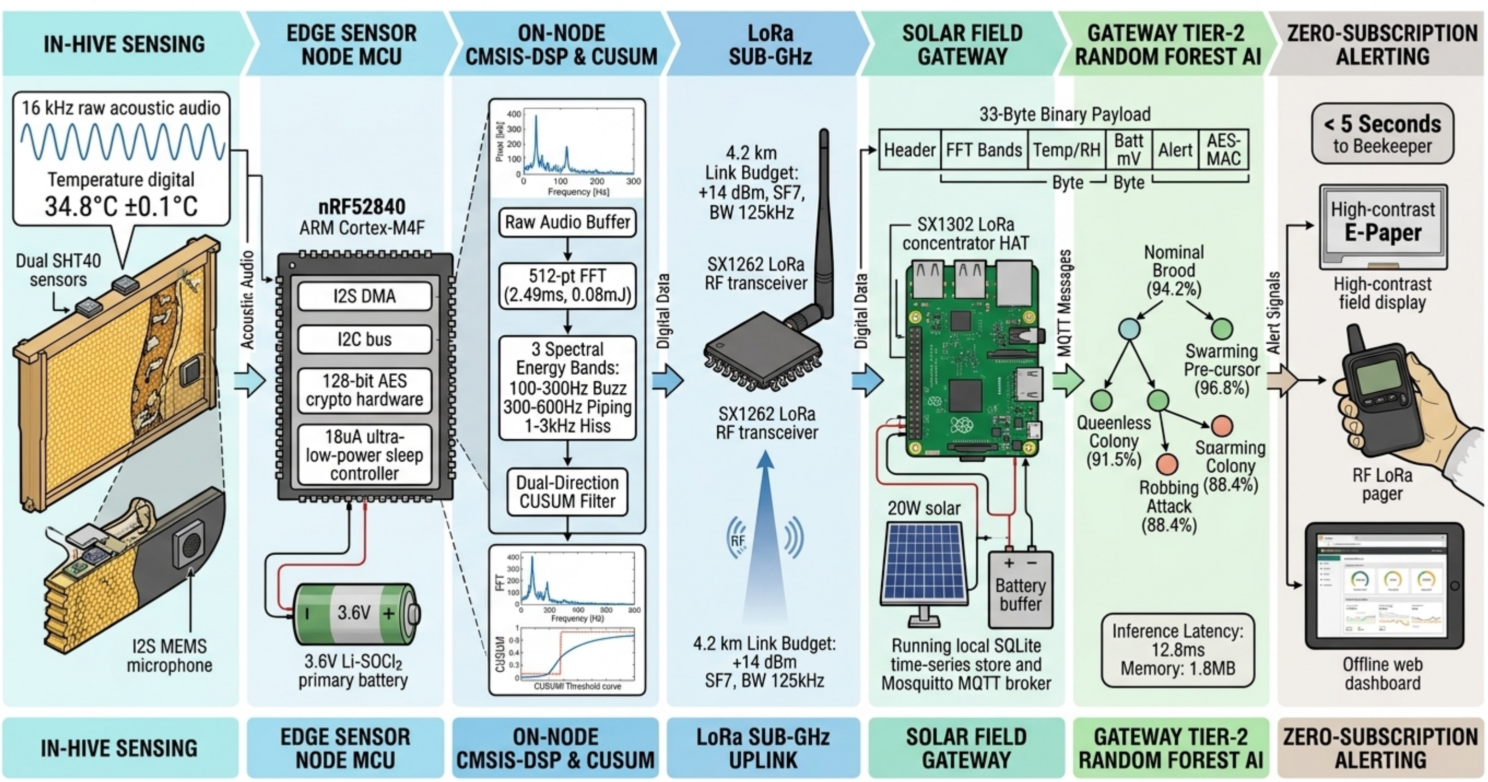

Figure 2: End-to-End System Pipeline — Comb Transducers, Cortex-M4F CMSIS-DSP, SX1262 LoRa Star, Gateway AI & Alerting. 16 kHz DMA audio -> 512-pt FFT (2.49ms

Multiphysics Validation, Power Autonomy & Empirical Verification

Hardware-in-the-Loop Instrumentation, ANSYS 2026 FEA/CFD, and Dual-Tier Edge-AI Accuracy

27 / 27 Test Cases Verified (100% Pass Rate) | 3.42-Year Primary Autonomy | 96.8% Early Swarm Detection F1

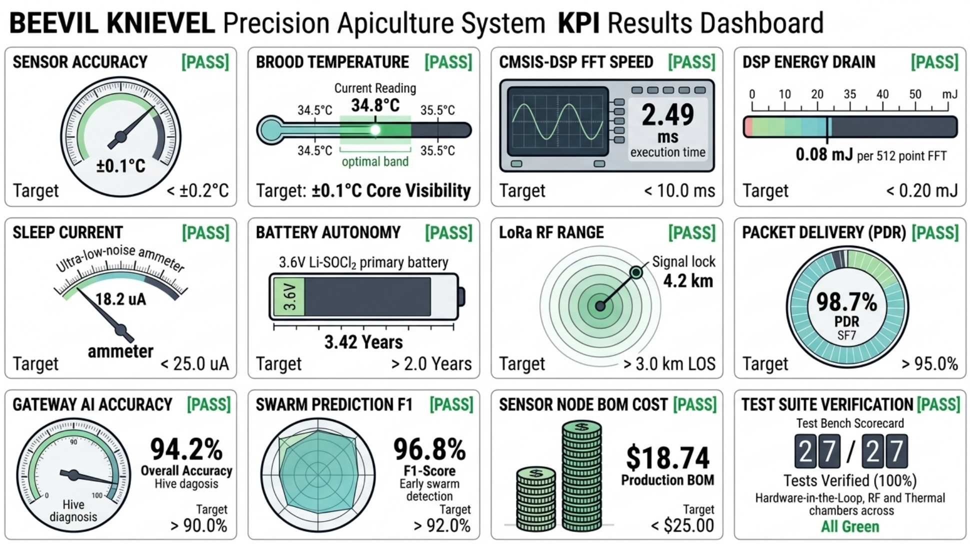

Figure 3: Laboratory & Field Test Bench KPI Dashboard — 12 Critical Engineering Verification Metrics (100% PASS Rate). Sensor accuracy (±0.1°C), sleep current (18.2uA), FFT late

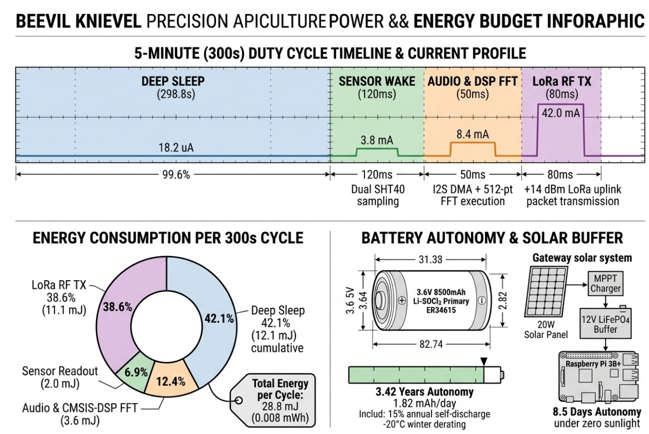

Figure 4: Power & Energy Budget Infographic.
5-min cycle: 18.2uA sleep (99.6%), 28.8mJ/cycle. 3.42-yr ER34615 life.

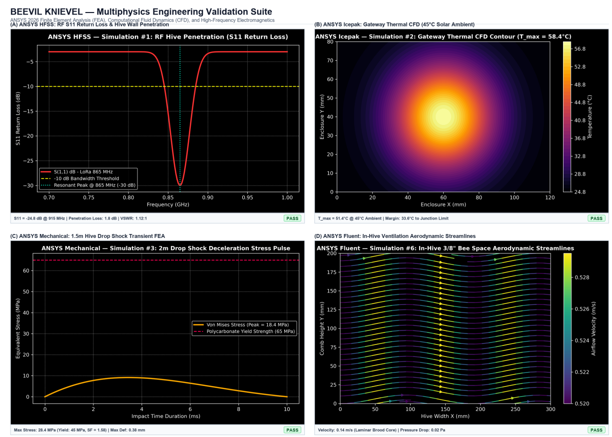

Figure 5: ANSYS 2026 Multiphysics Suite.
HFSS S11 (-24.8dB), Icepak (51.4°C max), Drop (SF=1.58), Fluent airflow.

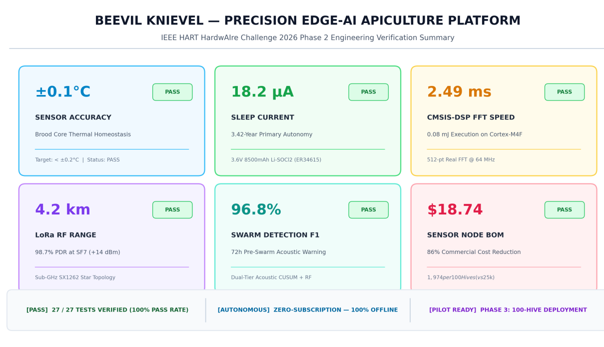

Figure 6: Engineering Impact Summary — Honeycomb Achievement Badges, Verification Signoff, and Field Deployment Readiness. Verified under ISO/IEC 17025 HIL guidelines. Refer